Yisheng Zhong

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Summary

Ph.D. student in Cybersecurity at **George Mason University** (advised by Dr. Zhuangdi Zhu). Research focuses on the security and privacy of large language models (LLMs), including **unlearning**, **safety alignment**, and **defenses against LLM-driven content extraction and misuse**. Previously earned an M.S. at the University of Chinese Academy of Sciences and conducted research on privacy-preserving federated learning at the State Key Laboratory of Information Security.

Research Interests: LLM Post-training (SFT, RLHF/DPO), Unlearning, Alignment.

Education

George Mason University, Ph.D. in Information Technology — **GPA: 4.00** | 2024 – Present

- Research on security and privacy of LLMs, with a particular interest in **LLM unlearning**.
- Graduate Research Assistant (2024); Graduate Teaching Assistant (2025).

University of Chinese Academy of Sciences, M.S. in Cyber Security — **GPA: 3.56** | 2021 – 2024

- Relevant coursework: Machine Learning, Deep Learning, Security Protocols, Applied Cryptography.
- Focus on privacy-preserving computing and defenses against adversarial attacks in machine learning.

Harbin University of Science and Technology, B.S. in Computer Science — **GPA: 3.84 (Top 1%)** | 2017 – 2021

- Relevant coursework: Data Structure, Probability & Statistics, Linear Algebra, Pattern Recognition.
- Received **direct admission offer** to pursue a master's degree at the Chinese Academy of Sciences.

Research Experience

1. **DUET: Distilled LLM Unlearning from an Efficiently Contextualized Teacher** *ICLR 2026 Accepted*
   Yisheng Zhong, Zhengbang Yang, Zhuangdi Zhu

   - Developed DUET, an LLM unlearning technique that (i) uses an efficiently contextualized teacher (prompt-conditioned) to **demonstrate refusals on undesirable knowledge** and (ii) **distills this behavior into a student model**, achieving targeted forgetting with minimal utility loss.
   - Introduced **Top-K logit alignment** in place of full-vocabulary KL, enabling more precise forgetting with better efficiency; on MUSE, average leakage decreased by 4% (ROUGE-Forget) and utility increased by 10% (ROUGE-Retain/MMLU), while training used about 1/645 of corpus tokens (2,233 vs. 1.44M).
   - Demonstrated robustness to reverse prompts and task-format shift (QA $\rightarrow$ continuation) on WMDP and MUSE; under reverse prompts, the in-context teacher's leakage rises from 4.52% to 37.62%, whereas DUET remains around 5.98%$\rightarrow$7.27%.

2. **Web Intellectual Property at Risk: Preventing Unauthorized Real-Time Retrieval by Large Language Models** *EMNLP 2025, Main Conference, Poster*
   Yisheng Zhong, Yizhu Wen, Junfeng Guo, Mehran Kafai, Heng Huang, Hanqing Guo, Zhuangdi Zhu

   - Designed a **semantic defense framework** that embeds optimized HTML policy cues to block LLMs from unauthorized real-time extraction and redistribution of web content.
   - Improved defense success rates from 2.5% to 88.6% across multiple proprietary LLMs, outperforming configuration-based defenses such as robots.txt.
   - Supported three granular protection goals: refusal to answer, partial masking, and redirection to the source, ensuring both autonomy and content discoverability.

3. **CATNIP: LLM Unlearning via Calibrated and Tokenized Negative Preference Alignment** *Under review, ICML 2026*
   Zhengbang Yang, Yisheng Zhong, Junyuan Hong, Zhuangdi Zhu
   - Propose CATNIP, a **retention-data-free unlearning framework** using adaptive reverse-policy calibration and token-level weighting to achieve precise knowledge removal without collateral damage to utility.
   - Demonstrate stronger forget–retain trade-offs than methods GA/NPO/SimNPO/FLAT on benchmark WMDP (Bio/Cyber) and MUSE-Books; remains robust with scarce, short QA-format unlearning data.
   - Ablations isolate gains from calibration and tokenization, showing both components are necessary to achieve effective forgetting without retention or contrastive pairs.
4. **Hierarchical Federated Unlearning for Large Language Models** *FedKDD 2025*
   Yisheng Zhong, Zhengbang Yang, Zhuangdi Zhu
   - Proposed **Federated Unlearning Merge (FULM)**, a scalable and privacy-preserving framework for decentralized LLM unlearning requests.
   - Decoupled forgetting and retention into dual task adapters and introduced a **hierarchical merging strategy** to mitigate inter- and intra-domain interference.
   - Outperformed baselines on TOFU (+8.9%) and WMDP+MUSE (+7.8%) benchmarks, demonstrating a superior trade-off between forgetting and retention.
5. **PROFL: A Privacy-Preserving Federated Learning Method with Stringent Defense Against Poisoning Attacks** *CSCWD 2023, Oral*
   Yisheng Zhong, Li-Ping Wang
   - Developed a **Byzantine-robust federated learning framework** combining similarity-based and statistical defenses against hidden data poisoning.
   - Integrated two-trapdoor Homomorphic Encryption for secure computation; outperformed comparable schemes in extreme cases by 13%–56%.

Competition Experience

Mathematical Contest in Modeling 2020 — *Meritorious Winner* (International First Prize) Team Leader

- Led the modeling of a 3D cellular automata system governed by differential equations and optimized via a genetic algorithm; oversaw the end-to-end execution from algorithm design to implementation.

Academic Services

Reviewer: ICLR 2024–2025, TIFS 2025

Honors & Awards

Meritorious Winner at the Mathematical Contest in Modeling (**First Prize**), 2020

Annual Scholarships of the Chinese Academy of Sciences, 2021–2023

First-Class Scholarships of Harbin University of Science and Technology, 2017–2021